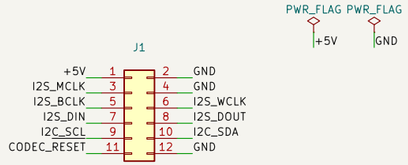
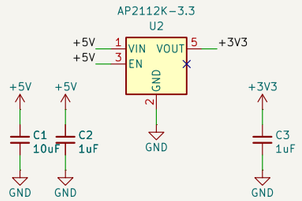
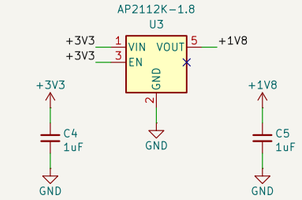
CAR / TELEMETRY INTERFACE + POWER



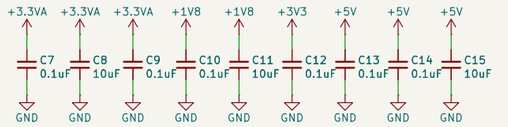
Telemetry / 5V in

I2S_DIN = audio TO helmet (codec input)
I2S_DOUT = mic audio FROM codec
Codec is I2S slave; BCLK can feed PLL
if MCLK is unavailable.



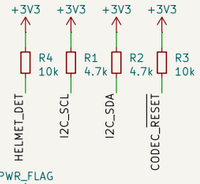


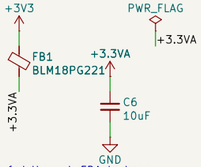
AUDIO CODEC (I2S audio, I2C control)



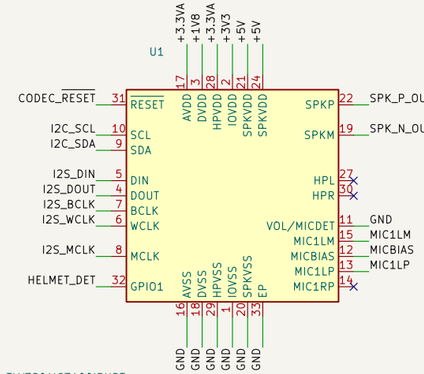
C7/C8 AVDD C9 HPVDD C10/C11 DVDD C12 IOVDD C13/C14/C15 SPKVDD
Place each 0.1uF within 2 mm of its supply pin.

I2C addr 0x18. Remove R1/R2 if the telemetry board already pulls up.





AVDD/HPVDD fed through FB1 to keep
digital + class-D switching noise
out of the mic preamp supply.



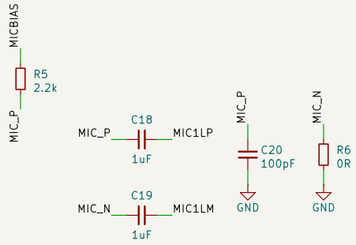
TLV320AIC3100IRHBT

MICBIAS = 2.0 V (CMR-2747PB-A rated Vs).
MIC PGA: MIC1LP/MIC1LM pseudo-differential.
Enable ADC HPF (-150 Hz) to cut wind rumble.
Cap speaker volume in firmware (hearing).
VOL/MICDET unused -> tied to GND.

SPEAKER EMI FILTER / MIC FRONT END / ESD

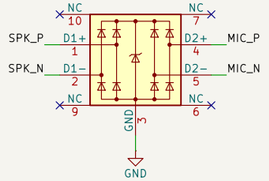


Filterless class-D -> FB + 1nF (TI EMI guidance)
before 1-2 m of harness. 2x 8R speakers in
series = 16R load (~0.7 W max @ 5 V).



R5 = mic load RL (2.2k per datasheet).
C20: RF shunt (radio TX near the cable).
R6: single mic-return-to-GND point, at codec.

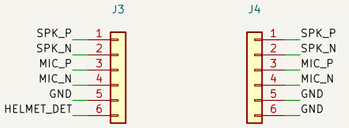
D1
TPD4E05U06DQAR





J2 pin 5 = cable shield/drain
(GND at board end only)

OFF-BOARD: cockpit harness + helmet (not placed on PCB, in BOM)

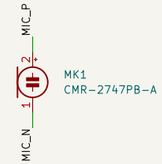


878-22-006-00-011101
878-20-006-00-012000

MAGNETIC BREAKAWAY: J3 (car, spring pins)
mates J4 (helmet pigtail, flat targets).
No latch -- pulls apart when driver exits.
J4 pin 6 strapped to pin 5 in the helmet:
HELMET_DET reads low when connected.







Speakers: in EPS ear pockets behind comfort liner.
Mic: chin bar, ~1 cm from lips, foam windscreen.
Cable: 4c + drain shielded, mic pair twisted.

Hadi Mirza
LV onboarding – driver radio audio (codec + breakaway helmet harness)

Formula Slug

Sheet: /
File: in-helmet-audio.kicad_sch

Title: In-Helmet Audio Interface

Size: B	Date: 2026-10-13	**Rev: 0.1**
KiCad E.D.A. 10.0.6		Id: 1/1